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24MA313

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Third Semester B.E. Degree: Test-I

Department of Mathematics

Engineering Mathematics-III

Duration: 1 Hour 15 Minutes | Max. Marks: 30

Instructions:

1. Answer all questions in **PART-A**.
2. Questions in **PART-B** have internal choice.

PART-A

Q.No.	CO	CD	PI	Marks
1	CO-1	1,3	1.1.1	07
	Given the equation of the regression lines: $4x - 5y + 33 = 0$ and $20x - 9y = 107$. Find the mean of x and y and the correlation coefficient. Find σ_y if $\sigma_x^2 = 9$.			
2	CO-2	1,3	1.1.1	08
	An urn A contains 2 white and 4 black balls. A second urn contains 5 white and 7 black balls. A ball is transferred from A to B and then a ball is drawn from B. Find the probability that it is white?			

PART-B

3 (a)	CO-1	1,3	1.1.1	08
	Fit a second degree parabola of the form $y = ax^2 + bx + c$ by the method of least squares for the following data:			

x :	1	2	3	4	5	6	7	8	9
y :	2	6	7	8	10	11	11	10	9

3 (b)	CO-2	1,3	1.1.1	07
	If A, B and C are independent events, show that the events: (i) $\overline{A}$ and $\overline{B}$ are also independent. (ii) $\overline{A}$ and B are also independent. (iii) A and $\overline{B}$ are also independent.			
	OR			

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4 (a)	CO-1	1,3	1.1.1	07
	Obtain the lines of regression and hence find the coefficient of correlation for the following data:			

x :	10	14	18	22	26	30
y :	18	12	24	6	30	36

4 (b)	CO-2	1,3	1.1.1	08
	Three students X Y and Z write an examination. Their chances of passing are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that (i) all of them pass (ii) at least one of them pass (iii) at least two of them pass.			

Course Outcome: At the end of the course the students will have the ability to	
CO-1:	Apply least square method to fit a curve for the given data and evaluate the correlation coefficient and regression lines for the data.
CO-2:	Determine the nature of the events and hence calculate the appropriate probabilities of the events.

Performance Indicators:

1.1.1	Apply mathematical techniques such as calculus, linear algebra, and statistics to solve problems.
1.3.1	Apply fundamental engineering concepts to solve engineering problems.

----- END -----

$9x = 5y - 93$

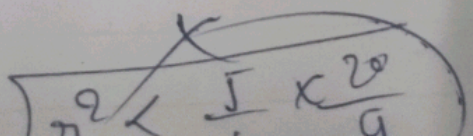
$x = \frac{5}{9}y - \frac{93}{9}$

$\frac{5}{9}$

$9y = 20x - 107$

$y = \frac{20}{9}x - \frac{107}{9}$

$\frac{20}{9}$



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JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Computer Science and Engineering

III Semester

Test I

Object Oriented Programming using C++

Duration: 1.15 Hr | Date: 10/11/2025 | Max. Marks: 30 Marks

NOTE: Question 1,4 & 5 are compulsory.
Answer any one between question 2 and question 3.

Q.No.	CO	CD	PI	QUESTION	MARKS
1a	CO1	L2	1.4.1	Explain the concept of inline functions with its advantages. Write a C++ program to calculate the sum of two numbers using inline function.	07
1b	CO1	L3	1.4.1	Write a C++ program to demonstrate function overloading by creating a function maxValue() that finds the largest value among: i) Two integers ii)Two floating-point numbers and iii) Three integers.	08
2a	CO2	L2	2.1.2	What is a constructor? Discuss the rules for defining constructors in C++. Write a suitable example to illustrate the use of a constructor.	05
2b	CO2	L3	2.1.2	Create a class *Sample* with two private integer data members. Use member function to read the data members. Write a C++ program to calculate and display the sum of these two members using a non-member function that is declared as a friend of the class.	05
				OR	
3a	CO2	L3	2.1.2	Write a C++ program to demonstrate the concept of reference variables by swapping two numbers using a function with reference parameters. Also explain how reference parameters differ from pointers or value parameters.	05
3b	CO2	L2	2.1.2	Explain default arguments with an example program.	05
4	CO1	L2	1.4.1	Explain any three key features of Object-Oriented Programming with suitable example.	03
5	CO2	L1	2.1.2	Differentiate between dynamic memory allocation and static memory allocation.	02

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Computer Science and Engineering

III Semester BE: **Test-1**

Digital System Design

Duration: 1 Hr. 15 Min. **Date**:11.11.2025 **Max. Marks**: 30

NOTE: Question No. 1 is compulsory, answer either Question No. 2 or 3 completely.

Q.NO	CO	CD	PI	QUESTION	MARKS
1(a).	1	L3	2.2.3	Using the Quine Mc Clusky method. Find the Prime Implicants and Essential Prime Implicants for the given Boolean function. F (A, B, C, D) = Σm(0, 1, 2, 8,10,11,14,15)	10
1(b).	1	L3	2.2.3	Simply the Boolean function using **Boolean laws and theorems** and Implement the results using **logic gates**. i. x'y'z+x'yz+xy' ii. A'B'C'+ A'BC'+ AB'C'+ ABC'	05
2(a).	1	L3	2.4.1	Simplify the Boolean function in **SOP** and **POS** for the given function: **f(p,q,r,s)= Σm(0,1,2,5,8,9,10)** using **K-MAP**. Implement the results using **logic gates**.	10
2(b).	1	L3	2.2.3	A Boolean function of 3 input variables x,y,z produces an output of 1 whenever 2 or more input variables have the value 1 else the output is 0. Construct the Truth table and express the function in canonical SOP form and canonical POS form.	05
				OR	
3(a).	1	L3	2.4.1	Simplify the following using K-MAP in **POS** for the given Boolean function: f(A,B,C.D)= πM(0,1,2,3,4,5,7)+dc(8,9,11,12,13,15) Implement using Universal gates.	10
3(b).	1	L2	1.3.1	What is Universal gates? Realize Basic gates(AND, OR, NOT) using Universal gates (NAND and NOR).	05

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JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Computer Science and Engineering

III Semester: Test I

Data Structures

Duration: 1hour 15 mins | Date: 11-11-2025 | Max. Marks: 30 Marks

NOTE: Question 1 is compulsory. Answer any one between question 2 and question 3.

Q.NO	CO	CD	PI	QUESTION	MARKS
1.	CO1	L2	1.4.1	a) Discuss how Dynamic Memory Management is implemented in C Programming Language with different functions and examples.	7
	CO2			b) Differentiate between Arrays and Linked list. c) Explain in brief the types of linked list.	4+4
2.	CO1	L3	2.1.2	a) Write a C program that defines a function called find ( ) that's returns 1 if the specified character is found in a given string else returns 0.	5+5+5
	CO1			b) Develop a C function that reverses the contents array of size 'n' using pointers and prints it.	
	CO2			c) Develop a function in C language called **insert(list, val)** where **list** is a pointer to the single linked list and **val** is the key value to be inserted. This function should insert **val** to the end of the list if **val** is not in the list and return the updated list. If **val** already exists, a suitable message to be displayed.	
				OR	
3.	CO1	L3	2.1.2	a) Write a C program that finds whether the given string is palindrome or not using pointers.	5+5+5
	CO1			b) Demonstrate the difference between structures and unions with suitable example.	
	CO2			c) Develop a function in C language called **search_update(list, X, Y)** where **list** is a pointer to the single linked list and **X and Y** are the key values. This function should search for **X** in the list and replace it with **Y** if **X** exists, otherwise a suitable message to be displayed.	

validity of the following argument

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Computer Science and Engineering

III Semester: CIE-1 / EVENT 1

COMPUTER ORGANIZATION

Duration (In hrs): 1.15 Hrs Date:12/11/2025 Max. Marks: 30

NOTE: Question 1 is compulsory, Question 2 and 3 has internal choice.

Q.No	CO	CD	PI	QUESTION	MARKS
1.	CO1	L3	1.4.1	a. Describe different types of addressing modes used in RISC architecture, giving one example for each b. Perform 2's complement addition on (-14) & (+9) c. Perform 2's complement subtraction on 27 & 13	8+3+4 = 15
2.	CO1	L2	2.3.1	a. Illustrate the basic structure of a computer using a neat block diagram and write a brief note on the functionality of each unit. b. Write an assembly level language code for sum of n numbers using branching and indirect addressing.	8+7 = 15
				OR	
3.	CO1	L2	2.3.1	a. Discuss the factors that influence the performance and speed of a computer system. b. Write an assembly program to find the sum of all scores for each test and store the respective totals in SUM1, SUM2, and SUM3	8+7 = 15

CO	Course Outcomes	Cognitive Domain
CO1	Apply the basic concepts of computer, instruction set architecture, function and characteristics of a computer system and design the assembly language for the sequence of machine instruction.	L3

PI	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.3.1	Able to apply computer engineering principles to formulate modules of a system with required applicability and performance

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Department of Computer Science and Engineering

UNIVERSAL HUMAN VALUES

III Semester B.E: Test 1

Duration: 45 Mins. Date: 12.11.2025 Max. Marks: 15

NOTE: Answer any 2 full questions.

Q.NO	CO	CD	QUESTION	Marks
1	CO-1	L2	Explain the concept of self-exploration and its significance in value education.	10
2	CO-1	L2	What is the purpose and motivation of the Value Education course? Explain with reference to Universal Human Values	5
3	CO-2	L2	Describe the coexistence of 'I' and Body in a human being.	5

Course Outcomes (COs)

CO	Course Outcomes	Cognitive Domain
CO1	Understand self (I) clearly through self-exploration and demonstrate natural acceptance in situations by right understanding	L2
CO2	Distinguish between self (I) body, and co-existence of both to be in harmony.	L2
CO3	Relate to family and society by understanding nine universal human values in human- human relationship for harmony and prosperity.	L2
CO4	Explain interconnectedness and mutual fulfilment at all four orders of nature for co- existence.	L2
CO5	Develop commitment for holistic understanding of harmony by natural acceptance of human values in his/her profession and society.	L2

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create